ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY
GUWAHATI

Course Structure and Syllabus

(From Academic Session 2018-19 onwards)

B.TECH
POWER ELECTRONICS AND INSTRUMENTATION ENGINEERING

7^{th} SEMESTER

ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY
Guwahati

Course Structure
(From Academic Session 2018-19 onwards)

B.Tech 7th Semester: Power Electronics and Instrumentation Engineering

Semester VII/B.TECH/PEI

Sl. No	Sub-Code	Subject	Hours per Week			Credit	Marks	
			L	T	P	C	CE	ESE
Theory								
1	PI181701	Power Electronics Device and Circuits	3	0	0	3	30	70
2	PI181702	Optical Instrumentation	3	0	0	3	30	70
3	PI1817PE2*	Program Elective-2	3	0	0	3	30	70
4	PI1817OE2*	Open Elective-2	3	0	0	3	30	70
5	HS181704	Principles of Management	3	0	0	3	30	70
Practical								
1	PI181711	Power Electronics Lab	0	0	2	1	15	35
2	PI181722	Project-1	0	0	6	3	50	50
3	SI181721	Internship-III (SAI - Industry)	0	0	0	2	-	200
TOTAL			15	0	8	**21**	**215**	**635**
Total Contact Hours per week: 23								
Total Credit: 21								

PROGRAM ELECTIVE – 2		
Sl.No.	**Subject Code**	**Subject**
1	PI1817PE21	Biomedical Instrumentation
2	EE1817PE32	Computer Networks
3	PI1817PE2*	Any other subject offered from time to time with the approval of the University

OPEN ELECTIVE – 2		
Sl.No.	**Subject Code**	**Subject**
1	PI1817OE21	Artificial Intelligence and Machine Learning
2	PI1817OE2*	Any other subject offered from time to time with the approval of the University

Detail Syllabus:

Course Code	Course Title	Hours per week L-T-P	Credit C
PI181701	Power Electronics Device and Circuits	3-0-0	3

Course Outcomes (COs):
At the end of the course, the students will be able to:

CO1: Understand the construction, principle of operation and the characteristics of Power semiconductor devices and fundamentals of thyristors and its family.

CO2: Analyze, operate and design ac-dc/dc-ac converters.

CO3: Analyze, operate and design dc-to-dc converters

CO4: Analyze and evaluate the operation of inverters, cycloconverters and AC controllers.

MODULE 1: Introduction

Power switching devices overview – Attributes of an ideal switch, application requirements, circuit symbols; Power handling capability – (SOA); Device selection strategy – On-state and switching losses – EMI due to switching - Power diodes - Types, forward and reverse characteristics, switching characteristics – rating.

MODULE 2: Controlled Devices

Current Controlled Devices: BJT's – Construction, static characteristics, switching characteristics; Thyristors – Physical and electrical principle underlying operating mode, Two transistor analogy – concept of latching; Gate and switching characteristics; converter grade and inverter grade and other types; series and parallel operation; --Voltage Controlled Devices: Power MOSFETs and IGBTs – Principle of voltage-controlled devices, construction, types, static and switching characteristics, steady state and dynamic models of MOSFET and IGBTs - Basics of GTO, MCT, FCT, RCT and GATT.

MODULE 3: Firing and Protecting Circuits

Necessity of isolation, pulse transformer, optocoupler – Gate drives circuit: SCR, MOSFET, IGBTs and base driving for power BJT. - Over voltage, over current and gate protections; Design of snubbers.

MODULE 4: DC to DC Converters

The chopper, Basic principle of DC chopper, Classification of DC choppers, Control strategies Basic DC-DC converter (switch regulator), topologies: Principle, operation and analysis for Step-down (Buck), Step-up (Boost), Step up/down (Buck-Boost), Continuous conduction and Discontinuous conduction operation Chopper configurations: Voltage Commutated, Current Commutated, Load Commutated Chopper Multi-phase chopper, Application of DC to DC converters.

MODULE 5: AC Voltage Controllers

Types of single-phase voltage controllers, single-phase voltage controller with R and RL type of loads.

MODULE 6: Inverters

Switched Single phase voltage source bridge inverters and their steady state analysis, modified Mc-Murray half bridge inverter, series inverters, three phase bridge inverters with 180^0 and 120^0 modes. single-phase PWM inverters, current source inverters, CSI with R load.

MODULE 7: Cycloconverters

Principles of operation, single phase to single phase step -up and step down Cycloconverters, three phase to single phase cycloconverters, output voltage equation for a cycloconverter.

Textbooks/Reference Books:

1. M D Singh and K B Khanchandani, “Power electronics”, TMH, New Delhi, 2nd ed., 2007.
2. M.H. Rashid, ‘Power Electronics - Circuits, Devices and Applications,’ Pearson Education India.
3. P.S. Bimbhra, ‘Power Electronics,’ Khanna Publishers.
4. Vedam Subramanyam, “Power Electronics – Devices, Converters and Applications”, New Age International Publishers Pvt. Ltd., Bangalore, 2nd ed. 2006.
5. Mohan, Undeland, and Robbin, ‘Power Electronics - Converters, Applications and Design,’ JohnWiley and Sons.
6. V.R.Moorthi, “Power Electronics”, Oxford University press, 2005.

Course Code	Course Title	Hours per week L-T-P	Credit C
PI181702	**Optical Instrumentation**	**3-0-0**	**3**

Course Outcomes (COs):
At the end of the course, the students will be able to

CO1: Acquire fundamental understanding of the basic solid-state semiconductor physics and optical fiber wave-guides.
CO2: Develop detailed knowledge of different types of optical sources and their applications.
CO3: Acquire understanding of different optical detectors.
CO4: To be able to design and development of optical sensors.

MODULE 1: Introduction to Optical Fibers

Ray theory, Optical fiber modes and Configuration-Single mode and multimode fibers, Step index and Graded index fiber, Fiber materials, Advantages of Optical Communication System. Transmission characteristics of optical fibers-Introduction, Attenuation, Material absorption loss, Fiber bend loss, Dispersion.

MODULE 2: Optical Fiber Sources & Connection

LED- Structure, Material, Direct and Indirect band gap materials, Characteristics, Power and Efficiency. Fiber Alignment, Fiber Splices, Fiber connectors.

MODULE 3: Laser

Absorption, Spontaneous and Stimulated Emission of radiation, Einstein relation, Population inversion, Threshold condition. Working of three and four level laser. Basic idea of solid state, semiconductors, gas and liquid laser. Basic concepts of Q-switching and mode locking. Laser applications for measurement of distance, velocity. Directional Coupler. LED vs LASER diodes.

MODULE 4: Optical Detectors

Optical detection principle, quantum efficiency, responsivity. PN junction photo diode, PIN photo diode, Avalanche photo diode, Noise in detectors, Photodiode materials.

MODULE 5: Optical Fiber Measurements

Measurements of Fiber Attenuation, Dispersion, Refractive Index Profile, Cut off wavelength, Numerical Aperature & Diameter.

MODULE 6: Optical Fiber Sensors

Introduction, Fiber optic sensor for industrial applications-Level, Flow, Displacement, Pressure, Temperature, Current sensor, Moire Fringe Modulation sensor, Laser Doppler Velocimeter.

Text/Reference Books:

1. C. K. Sarkar, D. C. Sarkar, "Optoelectronics and Fiber Optics Communication", New Age International Publishers.
2. John M. Senior, "Optical Fiber Communications", Pearson, 3rd Edition, 2010.
3. R. P. Khare, "Fiber optics and optoelectronics",Oxford University Press.
4. Gerd Keiser, "Optical Fiber Communications", McGraw Hill Education.
5. J. Gowar, "Optical Communication System", IEEE Press – 2nd Edition.
6. P. Bhattacharjee, "Semiconductor Optoelectronics", PHI.
7. Wilson and Hawkas, "Optoelectronics and Introduction", PHI.
8. A. Ghatak, K. Thyagarajan, "An Introduction to Fiber Optics", Cambridge University Press.
9. G. P. Agrawal, "Fiber optic Communication Systems", John Wiley & sons, NewYork, 1992

Course Code	Course Title	Hours per week L-T-P	Credit C
PI1817PE21	**Biomedical Instrumentation**	**3-0-0**	**3**

Course Outcomes (COs):

At the end of the course the students will be able to:

- **CO1:** Understand the components of medical instruments.
- **CO2:** Learn the electrical safety aspects and measurement errors in medical equipment.
- **CO3:** Gain the knowledge and functionality of medical equipment used for recording/measuring ECG and other cardiovascular parameters, Neonatology and drug delivery.
- **CO4:** Understand the function of general medical instruments.
- **CO5:** Perceive the principles, functions and design aspects of Pacemakers and defibrillators.

MODULE 1: Human Body & Subsystems

Brief description of neural. muscular. cardiovascular, nervous & respiratory system; their electrical. Mechanical and chemical activities.

MODULE 2: Transducers and Electrodes

Principles and classification of transducers for Bio-Electrode theory, Different types of electrodes-Hydrogen, Calomel, Ag-Agcl, pH, PO_2, PCO_2 electrodes. Selection criteria for transducers and electrodes.

MODULE 3: Electrophysiology and Bio-Potential Recording

The origin of Bio-potentials electrodes, biological amplifiers. ECG. EEG. EMG. PCG, EOG, lead systems and Measuring methods, typical waveforms and signal characteristics.

MODULE 4: Cardiovascular System Measurements

Measurement of blood pressure, blood flow, cardiac output, cardiac' rate, heart sounds, electrocardiograph, phonocardiograph, plethysmograph, echocardiograph Magneto-cardiograph.

MODULE 5: Measurement of Electrical Activities in Muscles & Brain

Electromyography: EEG & their interpretation, Respiratory system measurement, Measurement of gas volume & Flow rate, Carbon dioxide, Oxygen concentration, Respiratory controller.

MODULE 6: Medical Imaging

Diagnostic X-rays; MRI, thermography, ultrasonography. medical use of endoscopy.

MODULE 7: Care, Monitoring and Safety Measures

Elements of Intensive care measurement in basic hospital systems and components. Physiological effects of electric current shock hazards from electrical.

MODULE 8: Computer Applications and Biotelemetry

Real-time computer applications, data interpretation and processing, remote data recording and management.

MODULE 9: Therapeutic and Prosthetic Devices

Cardiac pacemakers, defibrillators, Design of pacemakers and defibrillators, ventilators, muscle stimulators, heart lung machine.

Text/Reference Books:

1. Khandpur R.S., Handbook of Biomedical Instrumentation, Tata McGraw Hill, 2016. 4.
2. John G.Webster, Medical Instrumentation: Application and Design, John Wiley and Sons Inc., 3 rd Ed., 2003.
3. Carr and Brown, Introduction to Biomedical equipment technology, 2011.

Course Code	Course Title	Hours per week L-T-P	Credit C
EE1817PE32	**Computer Networks**	**3-0-0**	**3**

COURSE OBJECTIVES:

- To teach the basics of the working of the Internet.
- To teach fundamentals of the protocols involved in the working of various layers of the Internet.

COURSE OUTCOMES:

At the end of this course, the students will be able:

CO1: To Understand the Basics of the Working of the Internet.

CO2: To Study Various Internet Protocols.

CO3: To Understand Various Security Measures in Internetworking.

CO4: To Understand Various Uses of Computer Networking.

MODULE 1: Introduction to Internet

a) To study Topology of Internet.
b) To study structure of Internet, Internet standards and Internet administration.
c) To study protocol layering.

MODULE 2: The Physical Layer

a) To learn Information Theory.
b) To learn coding of data, conversion methods between various types of data and multiplexing techniques.
c) To learn about various transmission media.
d) To learn about Circuit Switching, Packet Switching and various switches.

MODULE 3: The Data Link Layer

a) To learn error detection and correction techniques.
b) To learn DLC protocols.
c) To learn MAC protocols.
d) To learn various wired and wireless LAN protocols.
e) To learn mobile telephone protocols.
f) To learn satellite systems.

MODULE 4: **The Network Layer**

a) To learn packet switching methods.
b) To learn congestion control techniques.
c) To learn IP, IPv4, ICMPv4,
d) To learn Unicast Routing and Multicast Routing
e) To learn about IPv6 and ICMPv6,

MODULE 5: The Transport Layer

a) To learn about transmission layer issues protocols.
b) To learn about UDP, TCP-IP, SCTP

MODULE 6: The Application Layer

a) To learn basics of WWW, HTTP, FTP, E-Mail and DNS
b) To learn issues concerning multimedia in Internet.
c) To learn basics of Internet Security.

Textbooks/ Reference Books:

1. Computer Networking:: Stallings :: TMH
2. Computer Networks:: Tannenbaum :: PHI
3. Data Communication and Networking :: Forouzan :: McGraw Hill India

Course Code	Course Title	Hours per week L-T-P	Credit C
PI1817OE21	**Artificial Intelligence and Machine Learning**	**3-0-0**	**3**

Course Outcomes (COs):

At the end of the course the students will be able to:

- **CO1:** Understand Fundamentals of Modern AI.
- **CO2:** Analyse and understand the role of data in modern AI.
- **CO3:** Train and evaluate ML-based AI systems.
- **CO4:** Program AI systems using modern frameworks and programming tools such as Python notebooks.

MODULE 1:

Introduction to AI and ML, History of AI, Applications of AI & ML, Basics of ML (Classification, Regression, Clustering), Deep Learning Architectures, Artificial Neural Networks.

MODULE 2:

Mathematical Basics in Machine learning, Linear Algebra, Probability, Conditional Probability, Bayes' Theorem, Naive Bayes Classifier, Descriptive Statistics.

MODULE 3:

Introduction to Python, Introduction to modern programming tools such as Python notebooks, Data Types, Variables, Basic Input/ Output Operations, Conditional Execution, Loops, Logical and Bitwise Operations, String Manipulation. Functions, List, Tuples, Dictionaries.

MODULE 4:

Introduction to Python libraries, Introduction to Machine learning packages like Numpy, ScipY, Matplotlib and Pandas, Properties and their applications.

MODULE 5:

Types of learning in AIML, Ensemble Methods in Machine Learning: What are They and Why Use Them, Introduction to Tensor Flow and its applications, steps of building machine learning applications, Data Pre-processing for ML, Training and Testing of ML algorithms.

Text/Reference Books:

1. Python Programming-A Modular Approach with Graphics, Database, Mobile and Web Applications by Taneja Sheetal and Naveen Kumar
2. Core Python Programming - By Rao R. Nageswara.
3. Artificial Intelligence- Concepts and Applications by Lavika Goel.
4. Machine Learning Using Python by Pradhan Manaranjan
5. Machine Learning with Python for Everyone by Mark Fenner

Course Code	Course Title	Hours per week L-T-P	Credit C
HS181704	Principles of Management	3-0-0	3

MODULE 1: Introduction
Definition and meaning of management, Characteristics of management, importance of management, functions of management-planning, organising, directing, staffing, coordination and controlling etc., principles of management, Difference between administration and management

MODULE 2: Financial Management
Definition and management of financial planning, importance and characteristics of sound financial plan, concepts of capital- fixed capital and working capital, source of finance, fund flow statement.

MODULE 3: Marginal costing
Definition and meaning of marginal costing, advantages, marginal cost equation, contribution, profit-volume ratio, break even analysis, margin of safety.

MODULE 4: Cost Accounting
Cost Accounting- Concept and benefit, elements of cost, preparation of cost sheet with adjustment of raw materials, work-in-progress and finished goods.

MODULE 5: Capitalisation
Definition and meaning of capitalisation, over and under capitalisation.

MODULE 6: Motivation
Introductory observation, definition of motivation, motivational technique, features of sound motivational system.

MODULE 7: Leadership
Concept of leadership, principles of leadership, functions of leadership, qualities of leadership, different styles of leadership**.**

Textbooks/Reference Books:

1. Principle of Business Management: RK Sharma, Shashi K.Gupta
2. Business Organisation and Management: SS Sarkar, RK Sharma, Shashi K.Gupta
3. Industrial Organisation and Management: SK Basu, KC Sahu, B Rajviv
4. Principles of Management by Dr. A. K. Bora: Chandra Prakash, Guwahati.
5. Management Accounting: RK Sharma, Shashi K Gupta
6. Cost Accounting: SP Jain, K I Narang
7. Cost Accounting, RSN Pillai, V Bhagawati
8. Principles of Management: RN Gupta
9. Principles of Management: RSN Pillai, S. Kala
10. Principles of Management: Dipak Kumar Bhattacharjee

Course Code	Course Title	Hours per week L-T-P	Credit C
PI181711	**Power Electronics Lab**	**0-0-2**	**1**

Course Outcomes:

At the end of the course the students will be able to:

CO1: To expose students to operation and characteristics of power semiconductor devices and passive components, their practical application in power electronics.

CO2: To provide a practical exposure to operating principles, design and synthesis of different power electronic converters.

LIST OF EXPERIMENTS:

Exp1. Study the Static and dynamic characteristics of an SCR.

Exp2. Study the Output characteristics and transfer characteristics of Power MOSFET, DIAC, TRIAC.

Exp3. Design a Fan regulator using TRIAC.

.

Exp4. Design an ac-dc converter using line synchronized UJT triggering for firing the SCRs.

Exp 5. Design R and RC triggering scheme and determine the firing angle control range.

Exp 6. Simulate a step-down chopper for regulating the output voltage using time ratio control using PSIM.

Exp 7. Simulate a step-up chopper for regulating the output voltage using time ratio control using PSIM.

Exp 8. Simulate and design **s**peed control of DC separately excited motor with phase-controlled converter or DC-DC converter.

Course Code	Course Title	Hours per week L-T-P	Credit C
PI181722	**Project-1**	**0-0-6**	**3**
GUIDELINES WILL BE ISSUED BY THE UNIVERSITY FROM TIME TO TIME			

Course Code	Course Title	Hours per week L-T-P	Credit C
SI181721	**Internship-III (SAI - Industry)**	**0-0-0**	**2**
GUIDELINES WILL BE ISSUED BY THE UNIVERSITY FROM TIME TO TIME			

****